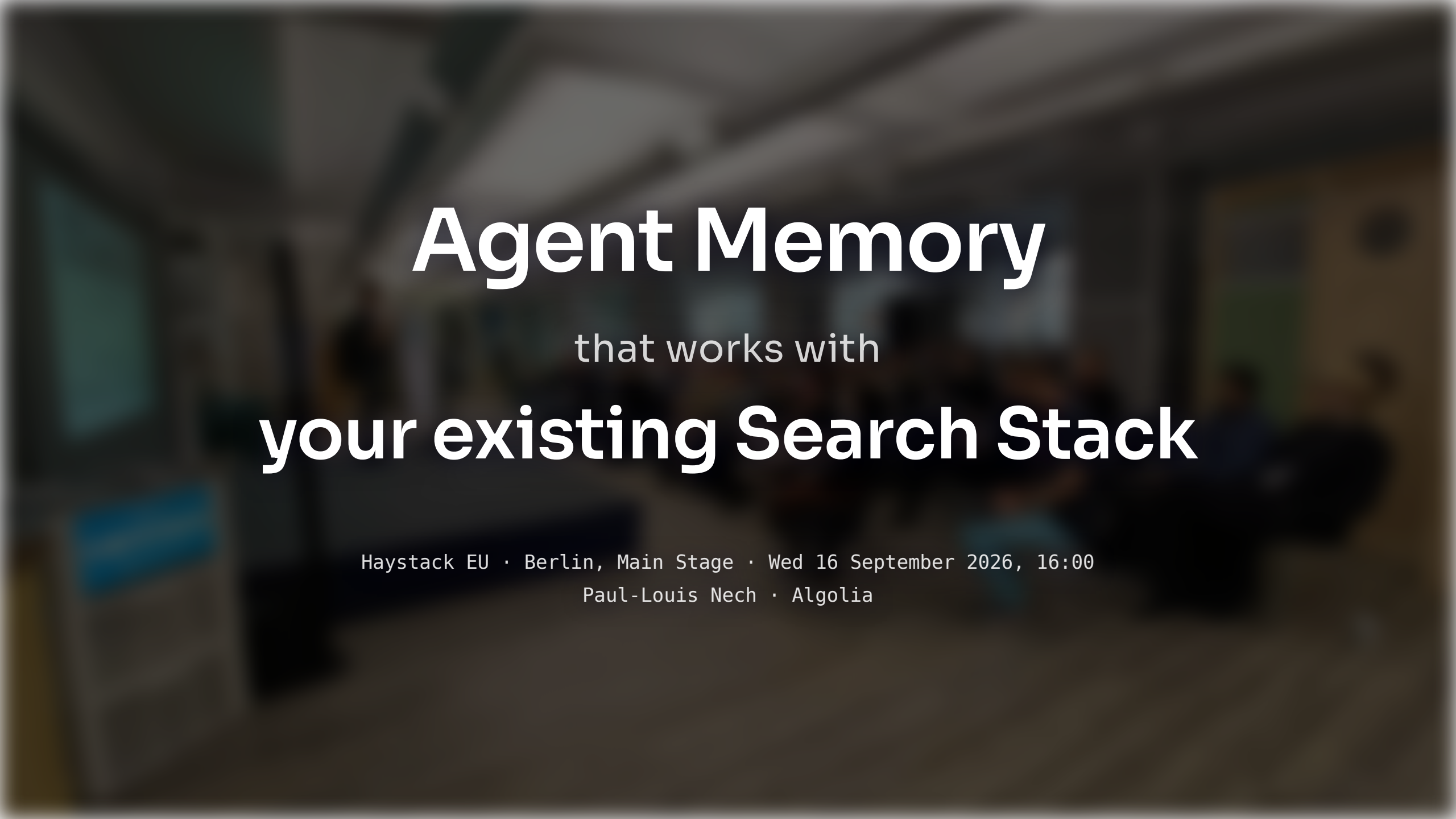




Agent Memory

that works with

your existing Search Stack

Haystack EU · Berlin, Main Stage · Wed 16 September 2026, 16:00

Paul-Louis Nech · Algolia

At least I remembered the interleaving tip.

Remember everything?

MEMORY

User prefers Vue.js over React
for new repos

technical

2026-08-12

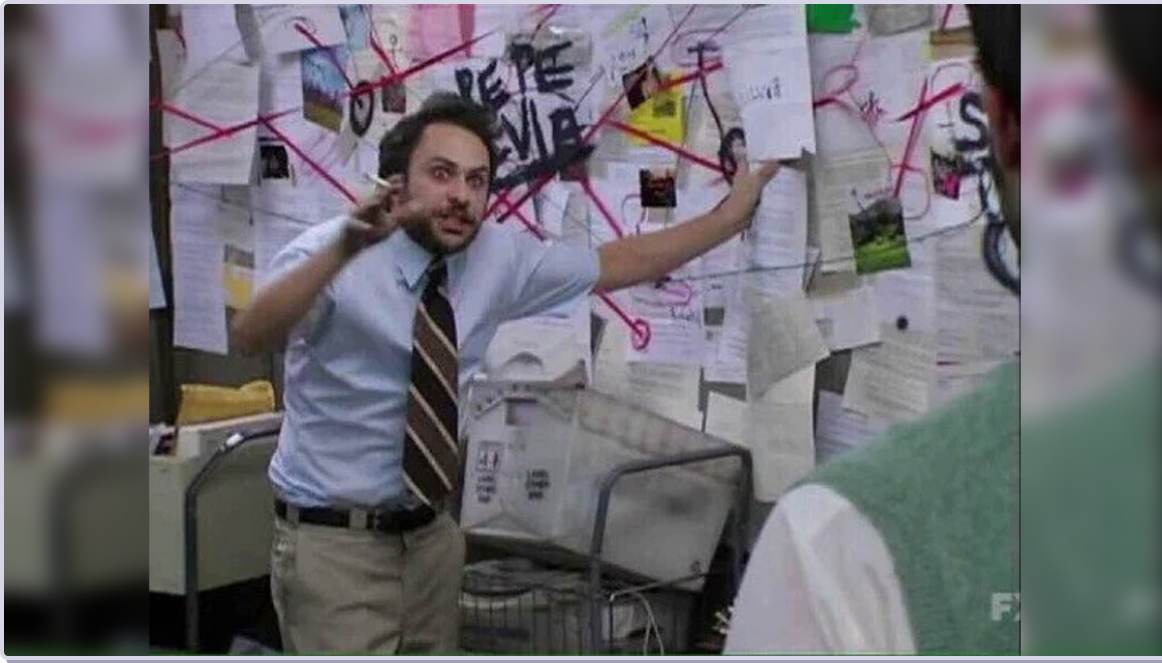


INTRO

This year I briefed my agents. Every. Single. Conversation.

Remembering not enough is bad.

The opposite is true too: bringing everything up is bad as well.



A CHATGPT SYSTEM PROMPT

```
[system]
...
model set context:
  2025-04-11: user prefers concise answers...
  2025-04-12: user is building a Vue app...
  2025-04-14: user lives in Paris...
  ... 40+ lines, every single turn ...
```

INTRO

**Remembering nothing is bad.
Remembering everything is bad too.**

False positives.

False negatives.

It's an F1 score. ✨

Memory is a Retrieval Problem

Haystack EU · Berlin, Main Stage · Wed 16 September 2026, 16:00

Paul-Louis Nech · Algolia

BTW hi 🖐️

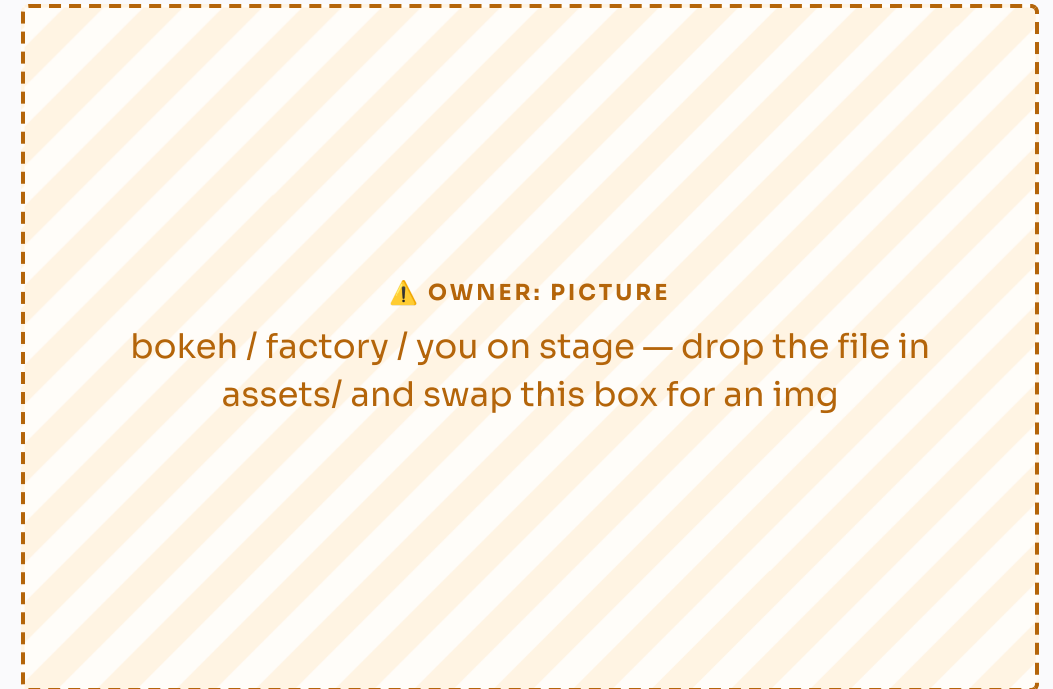
Paul-Louis Nech · Staff ML Engineer, Algolia @

We do Search.

(and Recommendations)

(and more)

Agent Studio is the
agentic harness on top.



Our turf

Our turf

6 min

Components

15 min

Side by side

5 min

Part 2 + close

7 min

**Agents are new.
Retrieval under constraints
is **our** turf.**

Budget 1: latency



Budget 2: tokens

estimated from our own wire bodies · ≈138 tokens per memory		
MEMORIES IN THE PROMPT	≈ TOKENS	× A 20-TURN CHAT
4	550	11k
10	1.4k	28k
30	4.2k	83k

You pay for every over-fetched memory, on every single turn.

Budget 3: attention

our own needle suite · 3 needles per document · April 2025, rerun September 2026		
	2025	2026
needle lost at	20k	never
128k costs you	a wrong answer	75–134 s

It used to be a wall. Now it's a bill.

Where benchmarks give you a baseline

thus a first impression



LongMemEval arXiv 2410.10813



LoCoMo arXiv 2402.17753

...and where your shopper doesn't care.

and has different success metrics

You already have query → results.

Components

Our turf
6 min

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5 min

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7 min

① The query.



four ways to build it

verbatim?

with topics?

saved triggers?

LLM rewrite?

COMPONENTS

Baseline: just the last N.

One conversation. Five queries.

TIRED	the last N memories	– no query at all
WIRED	the user, verbatim	"what haven't i tasted yet?"
+	with topics	... · topics: food, preferences
+	phrases saved at write time	recallTriggers: "coffee not tried yet"
HIRED	LLM rewrite, one round-trip	"specialty coffee Paris untried"

Same ask. The query is a design decision, not an inheritance.

The query you wrote at write time.

memory "... been to every specialty coffee bar on their Paris list
 except Substance in the 3rd ..."

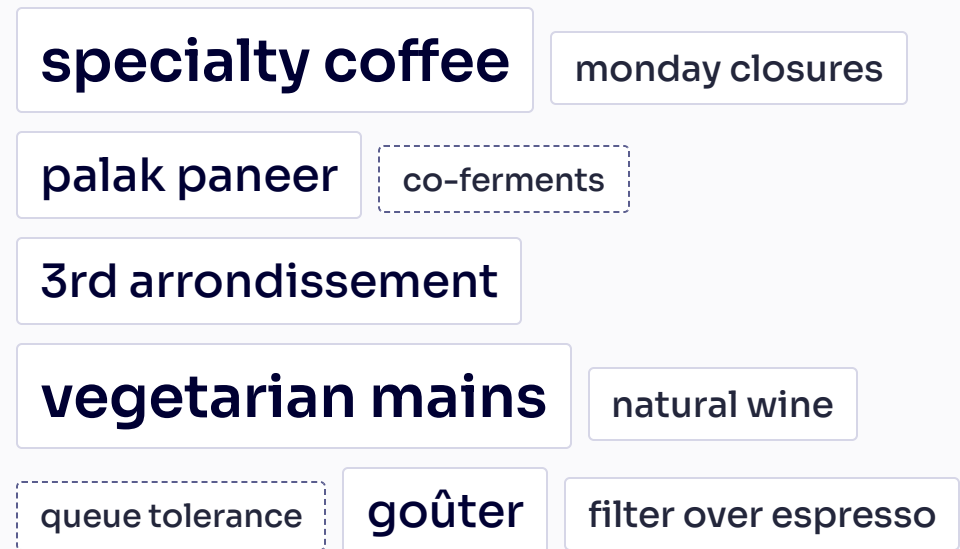
triggers "what coffee haven't i tried yet"
 "specialty coffee, 3rd arrondissement"
 "places still on my list"

Write the query when you write the memory.

...especially with topics.



closed: our real 18, one set, filterable



open: the corpus writes its own

COMPONENTS

② Retrieval.



levers you already tune

ranking formula

facets & filters

optional scoring

rules · synonyms

typo tolerance

📖 Docs

COMPONENTS

Same words: fine.
Same **meaning**: not so much.

recall@10 · keyword retrieval · n=1913 paired gold queries

the user's own words

81%

the same thing, rephrased

9.5%

same index. same records.

Three settings that moved recall.

keyword retrieval · n=1913 paired gold queries · the synthetic 670-memory bank	
SETTING	Δ RECALL@10
removeStopWords=false	-35.4 pp
searchableAttributes=[text]	-5.4 pp
optionalWords dropped	+2.9 pp
the other nine	noise

③ Formatting.



what you hand the model

attributesToRetrieve

snippeting

record format?

JSON? TOON? YAML? Markdown?

JSON

```
{"hits": [{"objectID": "d0cd1026e8ae51ef342eddaa108faeed2a1f6e2d0386dd1756a75e7fcfba34e2", "text": "User completed their undergraduate degree in Computer Science from UCLA.", "rawExtract": "I completed my undergrad in CS from UCLA, which has a great reputation in the industry.", "memoryType": "semantic", "createdAt": "1788961514", "keywords": "UCLA, Computer Science, undergraduate, bachelor degree..."}]}
```

TOON

```
hits[4]{objectID,text,rawExtract,memoryType,createdAt,keywords,topics,recallTriggers}:  
d0cd1026e8ae51ef342eddaa108faeed2a1f6e2d0386dd1756a75e7fcfba34e2,User completed their undergraduate degree in Computer Science from UCLA., "I co...
```

same 4 memories, ~11% fewer tokens

COMPONENTS

Measure it.

answer accuracy % · LongMemEval, one agent, four wire formats · n=60 per arm

json	53.3
toon	58.3 (+5.0)
yaml	56.7
markdown_kv	56.7

not significant

COMPONENTS

Query · Retrieval · Formatting.

COMPONENTS

Side by side

Our turf
6 min

Components
15 min

Side by side
5 min

Part 2 + close
7 min

SIDE BY SIDE

Same question. Three agents.

Four ways to remember.

our own stage rig · one question · September 2026 · tokens exact, clocks indicative			
	SECONDS	TOKENS IN	FACTS RECALLED
no memory	2.9	65	0
the last 5 turns	5.4	345	1
memory as a tool	18.1	9,184	1
preload + preflight	7.5	345	3

27× the tokens, to know less.

Part 2 + close

Our turf
6 min

Components
15 min

Side by side
5 min

Part 2 + close
7 min

PART 2 + CLOSE

**Some players go pure vector.
Sub-200 ms. Impressive!**

**First: make it work on
what you **already** have.**

Forgetting is retrieval too.

I asked the agent to compact its own memory.
91% of deletions
destroyed unique anchors.



[PART 2 + CLOSE](#)

Monday: measure it.

Review 5 conversations. Read 20 memories. Search twice yourself.

Retrieved, irrelevant?

Relevant, not retrieved?

Memory is a retrieval problem
— and retrieval is our domain.

RETRIEVED RECORDS · 7 · badges = the expected targets this record's own text contains

1 How does NeuralSearch rank results?	17700562-8481	neuralsearch	vector	keyword
2 How can I make sure Algolia NeuralSearch is working correctly?	15032552-8161	neuralsearch	keyword	documentation
3 Neural network vs. keyword search Algolia	en_253cf_0_2		keyword	
4 The role of semantically related keywords in search Algolia	en_90460_0_1		keyword	website
5 Algolia launches AI-powered Algolia NeuralSearch™ – The world's fastest, hyper-scalable, and cost-effective vector and keyword search API	en_81183_at_1	neuralsearch	vector	keyword website
6 Neural Search 101 Algolia	en_7cf43_0_1		no target	
7 Keyword search is built on natural language processing (NLP) Algolia	en_1056a_0_2		keyword	

• show 7 full hit objects

↳ unrecognised fields (not rendered above): answerText, clientToolCalls

(completion payload)

JUDGMENTS FOR THIS CASE · 1

0.77 PASSED relevance v2 RUBRIC v5 JUDGE deterministic SOURCE deterministic 12d ago

REASON (VERBATIM)

retrieval_recall=1.00 response_recall=0.60

RETRIEVAL RECALL 1.00 RESPONSE RECALL 0.60 axes

EXPECTED VS GOT derived in console

EXPECTED IN RETRIEVAL	neuralsearch	vector	keyword	website	documentation
EXPECTED IN ANSWER	neuralsearch ✓	vector ✓	keyword ✓	website ✗	documentation ✗

retrieval 5/5 · response 3/5 · score = $\sqrt{(1.00 \times 0.60)} = 0.77$

JUDGMENT METADATA

(payload)

Memory is just search.

THE QUESTION THE AGENT WAS ASKED

What is NeuralSearch?

019fc709...d70d

DIFFICULTY

easy

LANG

en

APPROVED

ANSWER KEY (CASEPAYLOAD.EXPECTED)

SHOULD CONTAIN CATEGORIES

Website

Documentation

KEYWORDS IN RESULTS

NeuralSearch

vector

keyword

(case payload)

RECEIPT · EXECUTION

MESSAGE 01a04946...e835

alg_msg_54ee15TcneElfXbv

CONVERSATION 01a04946...fc3d

EXECUTED 2026-08-28 16:49:28

(payload)

RECEIPT · JUDGING

JUDGMENT 01a04946...fd80

relevance

0.77

PASSED

JUDGED 2026-08-28 16:49:28

(payload)

WHAT THE AGENT DID · STORED COMPLETION

RESPONSE TEXT

NeuralSearch is Algolia's AI-powered hybrid search approach. It combines:

- **Keyword matching** for precise terms and exact wording

- **Semantic matching** to understand the concept or intent behind a query

- **Neural ranking** that combines keyword and semantic scores to determine relevance

It uses vectors and can learn from your own click and conversion events, helping results improve based on how users interact with your content. It's especially useful when users search naturally or use terms that don't exactly appear in your records.

NeuralSearch is less suited to use cases based primarily on exact matches, such as SKU-only searches.

Are you evaluating NeuralSearch for ecommerce, content, or another search experience?

TOOL CALLS · 1

algolia_search_index_algolia_www_prod_v2

arguments

RETRIEVED RECORDS · 7 · badges = the expected targets this record's own text contains

1

How does NeuralSearch rank results?

17700562...8481

neuralsearch

vector

keyword

Thanks.



These
slides

alg.li/haystack26

 PLN@algolia.com

 [@PLNech](https://github.com/PLNech)

 [Try Agent Studio](#)

[PART 2 + CLOSE](#)